

Zhiyao He

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Robot Learning · Reinforcement Learning · Vision-Language-Action Models · Robot Manipulation · Robotics Simulation

Profile

AI engineer building embodied AI systems that connect perception, language models, planning and control, from simulation (Gazebo, MuJoCo, Isaac Gym) to real robots. I aim to pursue a PhD in robot learning, focusing on how robots learn to perceive, reason, and act reliably in the real world.

Education

M.S. in Signal and Information Processing Hohai University, Nanjing, China Sep 2015 to Jul 2018

- Thesis on machine learning for runoff forecasting, leading to two peer-reviewed journal papers and one Chinese invention patent.

B.Eng. in Electrical Engineering and Automation Anhui Agricultural University, Hefei, China Sep 2011 to Jul 2014

- CGPA: 3.65 / 4.00. Completed the four-year program in three years, the only student in my cohort to do so.

A.S. in Computer Science University of the People, USA Nov 2021 to Nov 2024

- CGPA: 4.00 / 4.00

Standardized Tests

TOEFL: 5 / 6 (equivalent to 100 / 120), CEFR C1, Sep 2026

Publications and Patent

- **Zhiyao He**, Mingxiang Yang, Hao Wang, et al. "Annual Average Runoff Ensemble Forecast for Jinping I-Stage Hydropower Station Based on Elman Neural Network." Water Resources and Power, 2017.
- Chenming Li, Lei Zhu, **Zhiyao He**, et al. "Runoff Prediction Method Based on Adaptive Elman Neural Network." Water, 2019.
- **Zhiyao He** (Patent holder). "Medium- and Long-term Runoff Forecasting Method Based on Mutual Information, Kernel PCA, and Elman Neural Network." Chinese National Invention Patent, CN107463993B, granted 2020.

Selected Projects

Voice-Controlled Navigation and Manipulation on Unitree Go2-W team project with Macau Millennium College

- **Perception (my work)**: fine-tuned YOLO11 by transfer learning to detect a target bottle with a wrist-mounted depth camera, and mapped detections from 2D to 3D to locate it in the world frame.
- **Manipulation (my work)**: built a MoveIt 2 grasping pipeline (KDL inverse kinematics, OMPL/RRTConnect planning, FCL collision checking) that plans a collision-free path and executes the grasp.
- **Integration**: teammates built SLAM navigation and the ASR/LLM/OpenClaw/TTS pipeline; we defined the interfaces and integrated the full system in ROS 2, tested in MuJoCo simulation and on the real robot.

Retail Voice Agent on AgiBot Lingxi X2 industry project for CTM (Macau Telecom)

- In-store assistant that answers customer questions about products, plans, and pricing by voice, built in an offline and an online version.
- **Offline version**: runs on a Jetson for sites with unreliable network. A bridge node streams the X2's microphone audio to a local FastAPI backend running FunASR/Paraformer-zh ASR, Qwen2.5-7B via Ollama, pgvector RAG over a PostgreSQL product database, and Piper TTS.
- **Online version**: built with the Lingxi AgentsSDK, using DeepSeek as the LLM and MiniMax TTS with a Cantonese voice for Macau customers. It draws on three knowledge sources: the Lingxi platform knowledge base, the locally deployed RAG system, and third-party business systems such as CTM's.

Work Experience

AI Engineer WellWin Technology Co., Ltd., Hengqin & Macau, China Mar 2026 to Present

- Built the perception and grasping pipelines for voice-controlled navigation and manipulation on the Unitree Go2-W quadruped with a D1 arm (see Projects).
- Built offline and online voice agents for the AgiBot Lingxi X2 (see Projects), used in CTM retail stores and at live events with unstable network.

- AKI humanoid (team reproduction of Unitree's open-source Qmini): brought up and debugged the control and power systems, and ran RL locomotion training in Isaac Gym.
- Set up simulation environments and ran locomotion training for the Unitree G1/R1 humanoids in Gazebo, MuJoCo and Isaac Gym.

Independent Study and Research Nanjing, China

Jul 2019 to Feb 2026

- Completed an A.S. in Computer Science at University of the People (2021 to 2024, GPA 4.00/4.00) to strengthen my foundation in computer science and AI.
- Second journal paper published (Water, 2019); invention patent granted (2020).
- Admitted to the M.Sc. in Data Science, Trinity College Dublin (2025 intake).

Software Engineer Huawei Technologies Co., Ltd., Nanjing R&D Institute, China

Jul 2018 to Jun 2019

- Intelligent security systems team (face and vehicle recognition products): system development, deployment, and feature validation.
- Cross-team debugging and system-level testing.

Research Intern China Institute of Water Resources and Hydropower Research, Beijing, China

Jun 2016 to Jun 2017

- Built an end-to-end runoff prediction pipeline (data cleaning, feature engineering, model development, tuning, evaluation) that led to two journal papers and one invention patent.

Teaching and Outreach

- **Humanoid Robotics Curriculum, WellWin:** designed a 20-lesson curriculum for Macao secondary schools on the AgiBot Lingxi X2, progressing from no-code tools to ROS 2 with the AimDK SDK, with four real-robot capstones.
- **MYAIR 2026 (Macao Youth AI and Robotics Competition), Macao Science Center:** designed the humanoid-robot challenge and authored its rulebook: robots use vision to autonomously shoot a ball into the target cell of a QR-coded 3×3 grid.

Technical Skills

- Programming & Systems: Python, C++, Linux, Docker, Git
- Robotics: ROS/ROS 2, MoveIt 2, Nav2, Gazebo, MuJoCo, Isaac Gym, SLAM, Navigation, Perception
- AI & Robot Learning: PyTorch, Machine Learning, Computer Vision, Reinforcement Learning, LLMs, RAG, VLA
- Robot Platforms: Unitree G1/R1, Unitree Go2/Go2-W, AgiBot Lingxi X2
- Certificate: Deep Learning Specialization (DeepLearning.AI)

Honors and Awards

- Outstanding Graduate Student, Hohai University | 2017
- Hohai University Scholarship | 2015 to 2017
- Anhui Agricultural University Scholarship | 2012
- Outstanding Contribution Award, Huawei Nanjing R&D Institute | 2019
- President's List, University of the People (12 terms) | 2021 to 2024